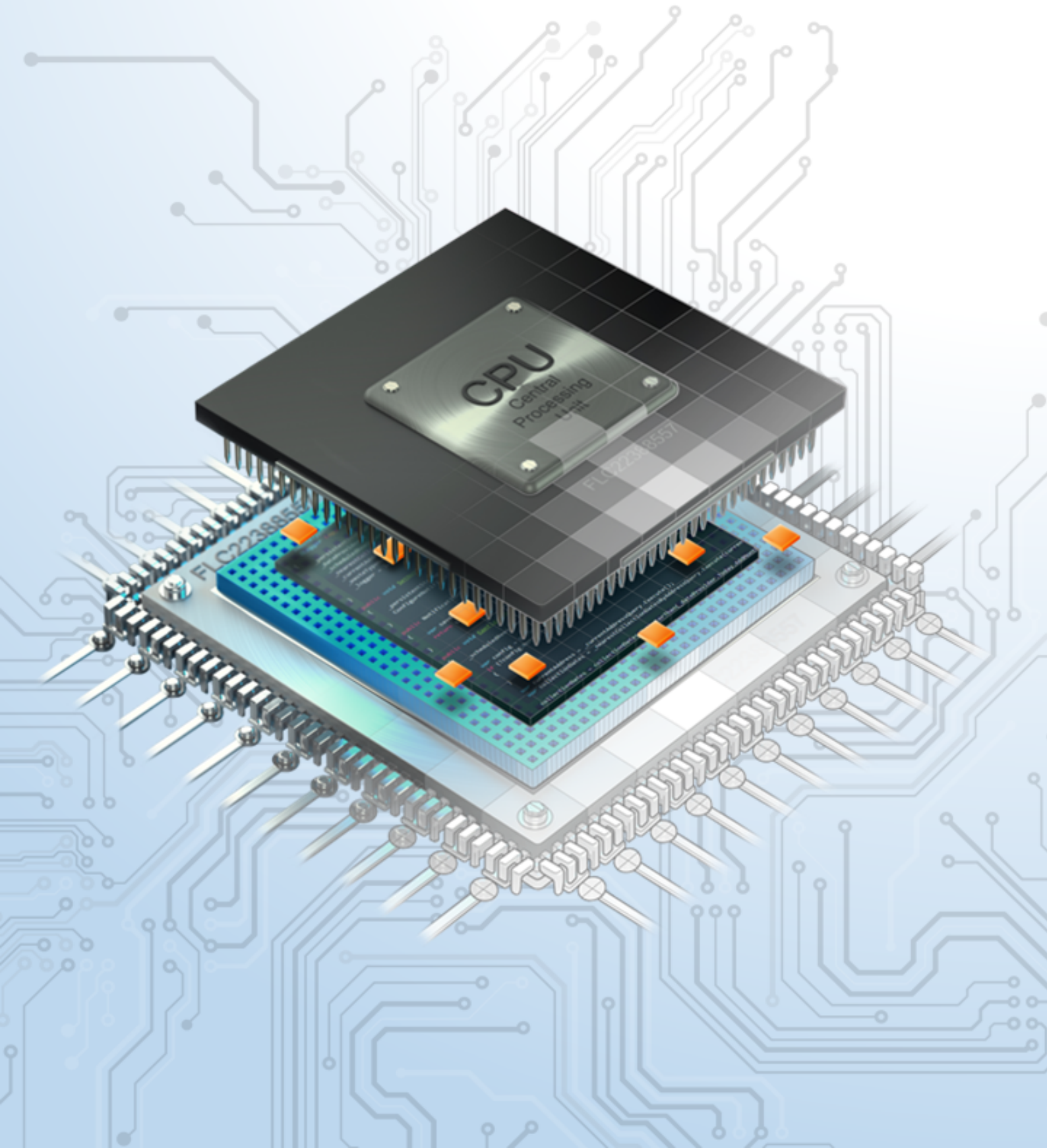




HO CHI MINH CITY UNIVERSITY OF TECHNOLOGY
COMPUTER ENGINEERING

Microcontroller





ĐẠI HỌC BÁCH KHOA - ĐHQG TP.HCM

KHOA KHOA HỌC & KỸ THUẬT MÁY TÍNH

BÁO CÁO THÍ NGHIỆM

VI ĐIỀU KHIỂN

LAB 1

LED

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TP. HỒ CHÍ MINH, 2026

Mục lục

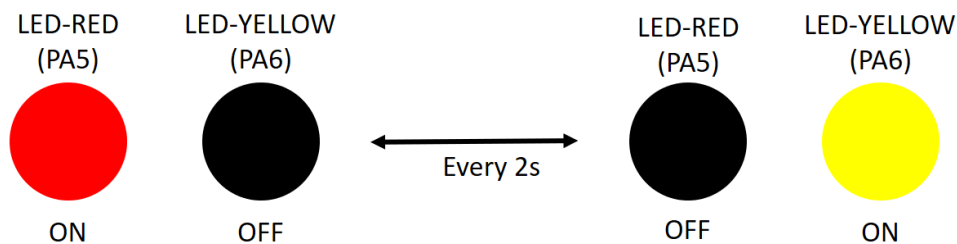
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1 REPORT

1.1 Exercise 1

From the simulation on Proteus, one more LED is connected to pin **PA6** of the STM32 (negative pin of the LED is connected to PA6). The component suggested in this exercise is **LED-YELLOW**, which can be found from the device list.

In this exercise, the status of two LEDs are switched every 2 seconds, as demonstrated in the figure bellow.



Hình 1: State transitions for 2 LEDs

Report 1: Depict the schematic from Proteus simulation in this report. The caption of the figure is a downloadable link to the Proteus project file (e.g. a github link).

Report 2: Present the source code in the infinite loop while of your project. If a user-defined functions is used, it is required to present in this part. A brief description can be added for this function (e.g. using comments). A template to present your source code is presented bellow.

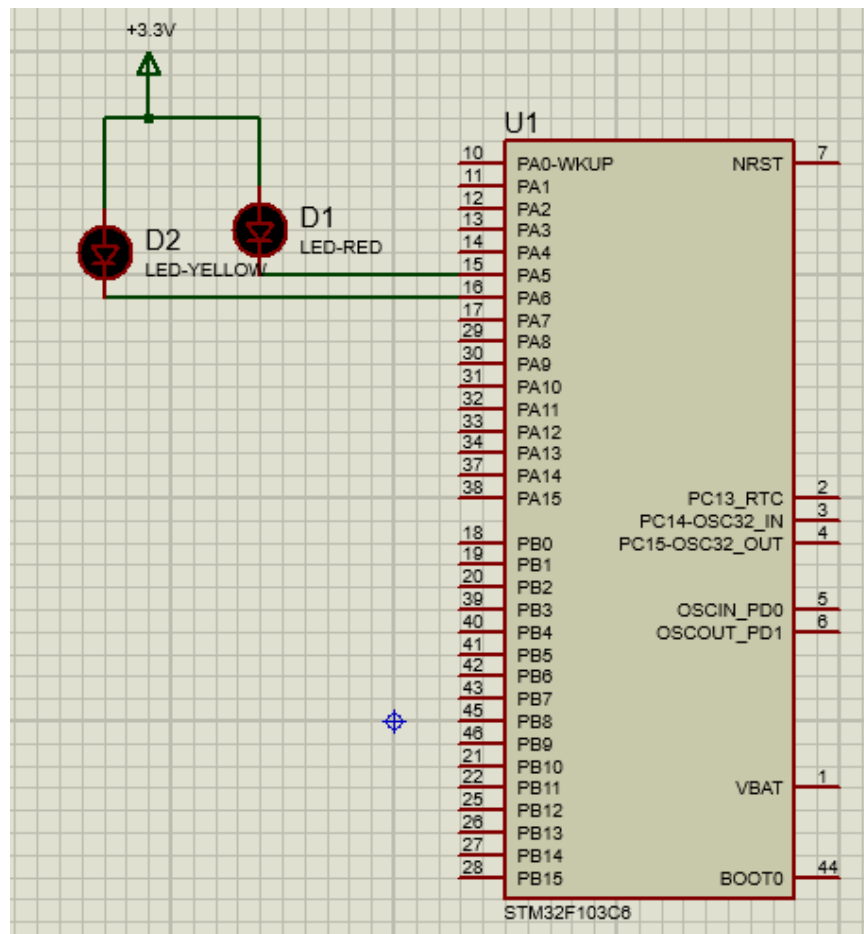
```
1 while (1) {  
2     // Switch state of two LEDs every 2 seconds  
3     HAL_GPIO_TogglePin(GPIOA, GPIO_PIN_5);  
4     HAL_GPIO_TogglePin(GPIOA, GPIO_PIN_6);  
5     HAL_Delay(2000);  
6 }
```

Program 1: Source code inside while loop for Exercise 1

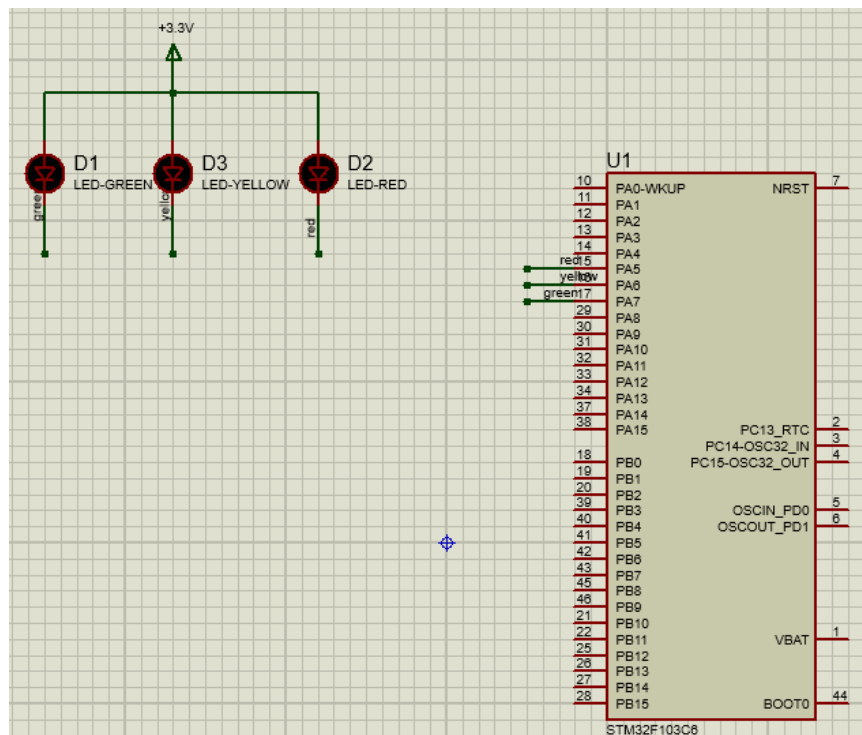
1.2 Exercise 2

Extend the first exercise to simulate the behavior of a traffic light. A third LED, named **LED-GREEN** is added to the system, which is connected to **PA7**. A cycle in this traffic light is 5 seconds for the RED, 2 seconds for the YELLOW and 3 seconds for the GREEN. The LED-GREEN is also controlled by its negative pin.

Similarly, the report in this exercise includes the schematic of your circuit and a your source code in the while loop.



Hình 2: Proteus Schematic for Exercise 1. Download project file at: [Excercise1/schematic/EX1.pdsprj](#)



Hình 3: Proteus Schematic for Exercise 2. Download project file at: Exercice2/schematic/EX2.pdsprj

Report 1: Present the schematic.

Report 2: Present the source code in while.

```

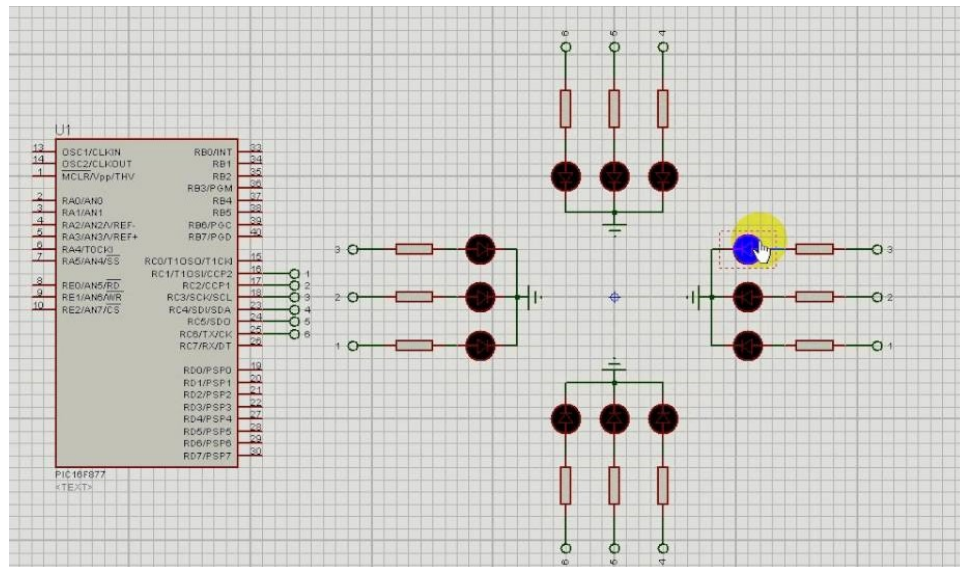
1 while (1) {
2     // Green LED ON (PA7 active low, RED & YELLOW OFF)
3     HAL_GPIO_WritePin(GPIOA, GPIO_PIN_5, GPIO_PIN_SET);
4     HAL_GPIO_WritePin(GPIOA, GPIO_PIN_6, GPIO_PIN_SET);
5     HAL_GPIO_WritePin(GPIOA, GPIO_PIN_7, GPIO_PIN_RESET);
6     HAL_Delay(3000);
7
8     // Yellow LED ON (PA6 active low, RED & GREEN OFF)
9     HAL_GPIO_WritePin(GPIOA, GPIO_PIN_5, GPIO_PIN_SET);
10    HAL_GPIO_WritePin(GPIOA, GPIO_PIN_6, GPIO_PIN_RESET);
11    HAL_GPIO_WritePin(GPIOA, GPIO_PIN_7, GPIO_PIN_SET);
12    HAL_Delay(2000);
13
14    // Red LED ON (PA5 active low, YELLOW & GREEN OFF)
15    HAL_GPIO_WritePin(GPIOA, GPIO_PIN_5, GPIO_PIN_RESET);
16    HAL_GPIO_WritePin(GPIOA, GPIO_PIN_6, GPIO_PIN_SET);
17    HAL_GPIO_WritePin(GPIOA, GPIO_PIN_7, GPIO_PIN_SET);
18    HAL_Delay(5000);
19 }

```

Program 2: Source code inside while loop for Exercise 2

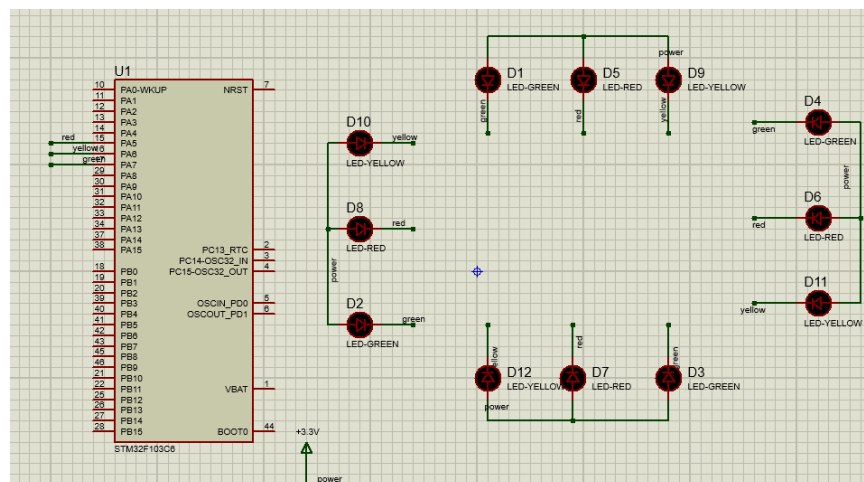
1.3 Exercise 3

Extend to the 4-way traffic light. Arrange 12 LEDs in a nice shape to simulate the behaviors of a traffic light. A reference design can be found in the figure bellow.



Hình 4: Reference design for a 4 way traffic light

Report 1: Present the schematic.



Hình 5: Proteus Schematic for Exercise 3. Download project file at: Excercise3/schematic

Report 2: Present the source code in while.

```
1 while (1) {
2     // Lane 1: GREEN (3s), Lane 2: RED (5s total)
3     HAL_GPIO_WritePin(GPIOA, GPIO_PIN_5, GPIO_PIN_SET);
4     HAL_GPIO_WritePin(GPIOA, GPIO_PIN_6, GPIO_PIN_SET);
5     HAL_GPIO_WritePin(GPIOA, GPIO_PIN_7, GPIO_PIN_RESET);
6     HAL_GPIO_WritePin(GPIOA, GPIO_PIN_8, GPIO_PIN_RESET);
7     HAL_GPIO_WritePin(GPIOA, GPIO_PIN_9, GPIO_PIN_SET);
8     HAL_GPIO_WritePin(GPIOA, GPIO_PIN_10, GPIO_PIN_SET);
```

```

9      HAL_Delay(3000);
10
11     // Lane 1: YELLOW (2s), Lane 2: RED (cont.)
12     HAL_GPIO_WritePin(GPIOA, GPIO_PIN_5, GPIO_PIN_SET);
13     HAL_GPIO_WritePin(GPIOA, GPIO_PIN_6, GPIO_PIN_RESET);
14     HAL_GPIO_WritePin(GPIOA, GPIO_PIN_7, GPIO_PIN_SET);
15     HAL_Delay(2000);
16
17     // Lane 1: RED (5s total), Lane 2: GREEN (3s)
18     HAL_GPIO_WritePin(GPIOA, GPIO_PIN_5, GPIO_PIN_RESET);
19     HAL_GPIO_WritePin(GPIOA, GPIO_PIN_6, GPIO_PIN_SET);
20     HAL_GPIO_WritePin(GPIOA, GPIO_PIN_7, GPIO_PIN_SET);
21     HAL_GPIO_WritePin(GPIOA, GPIO_PIN_8, GPIO_PIN_SET);
22     HAL_GPIO_WritePin(GPIOA, GPIO_PIN_9, GPIO_PIN_SET);
23     HAL_GPIO_WritePin(GPIOA, GPIO_PIN_10, GPIO_PIN_RESET);
24     HAL_Delay(3000);
25
26     // Lane 1: RED (cont.), Lane 2: YELLOW (2s)
27     HAL_GPIO_WritePin(GPIOA, GPIO_PIN_9, GPIO_PIN_RESET);
28     HAL_GPIO_WritePin(GPIOA, GPIO_PIN_10, GPIO_PIN_SET);
29     HAL_Delay(2000);
30 }

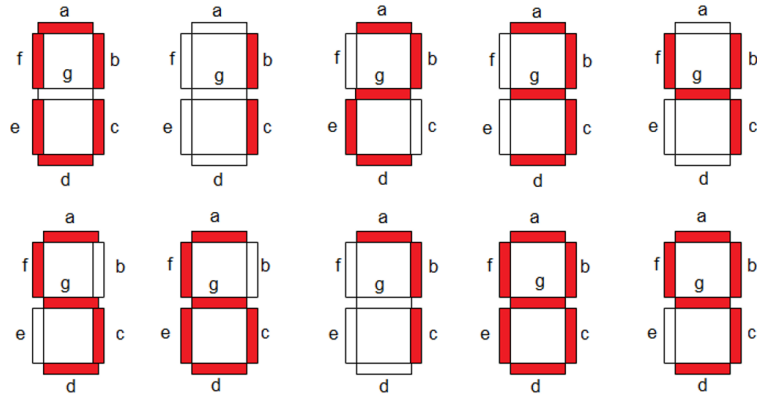
```

Program 3: Source code inside while loop for Exercise 3

1.4 Exercise 4

Add **only one 7 led segment** to the schematic in Exercise 3. This component can be found in Proteus by the keyword **7SEG-COM-ANODE**. For this device, the common pin should be connected to the power supply and other pins are supposed to be connected to PB0 to PB6. Therefore, to turn-on a segment in this 7SEG, the STM32 pin should be in logic 0 (0V).

Implement a function named **display7SEG(int num)**. The input for this function is from 0 to 9 and the outputs are listed as following:



Hình 6: Display a number on 7 segment LED

This function is invoked in the while loop for testing as following:

```

1 int counter = 0;
2 while (1){
3     if(counter >= 10) counter = 0;
4     display7SEG(counter++);
5     HAL_Delay(1000);
6 }

```

Program 4: An example for your source code

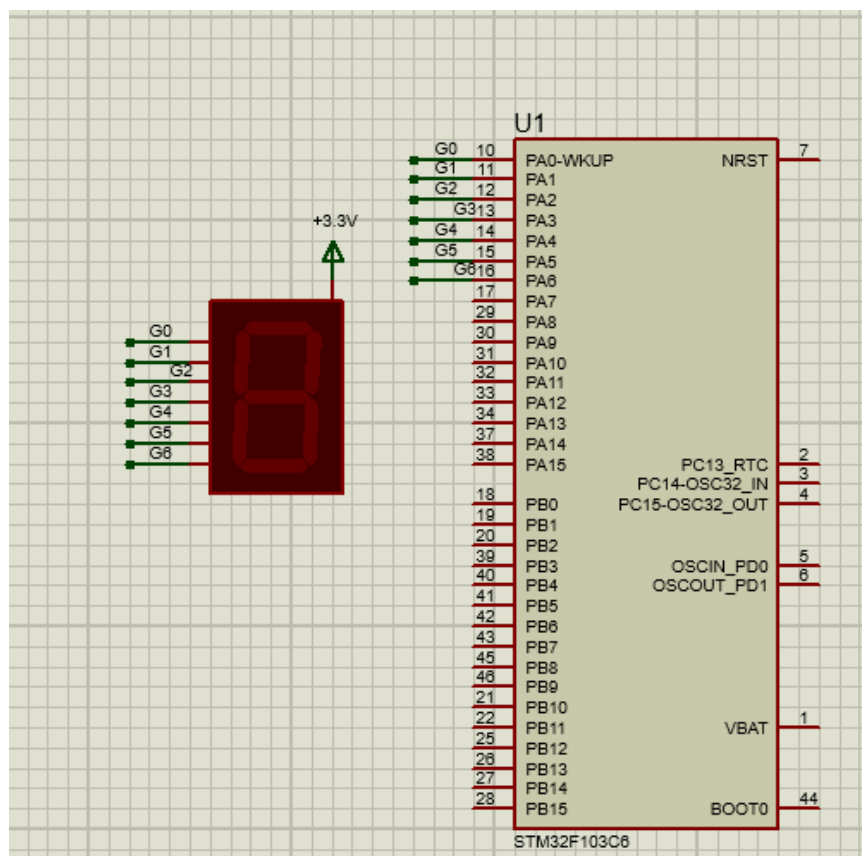
Report 1: Present the schematic.

Report 2: Present the source code for display7SEG function.

```

1 void display_7SEG(int count) {
2     switch(count) {
3         case 0:
4             HAL_GPIO_WritePin(GPIOA, GPIO_PIN_0, 0);
5             HAL_GPIO_WritePin(GPIOA, GPIO_PIN_1, 0);
6             HAL_GPIO_WritePin(GPIOA, GPIO_PIN_2, 0);
7             HAL_GPIO_WritePin(GPIOA, GPIO_PIN_3, 0);
8             HAL_GPIO_WritePin(GPIOA, GPIO_PIN_4, 0);
9             HAL_GPIO_WritePin(GPIOA, GPIO_PIN_5, 0);
10            HAL_GPIO_WritePin(GPIOA, GPIO_PIN_6, 1);
11            break;
12        case 1:
13            HAL_GPIO_WritePin(GPIOA, GPIO_PIN_0, 1);
14            HAL_GPIO_WritePin(GPIOA, GPIO_PIN_1, 0);
15            HAL_GPIO_WritePin(GPIOA, GPIO_PIN_2, 0);
16            HAL_GPIO_WritePin(GPIOA, GPIO_PIN_3, 1);
17            HAL_GPIO_WritePin(GPIOA, GPIO_PIN_4, 1);
18            HAL_GPIO_WritePin(GPIOA, GPIO_PIN_5, 1);
19            HAL_GPIO_WritePin(GPIOA, GPIO_PIN_6, 1);
20            break;
21        case 2:
22            HAL_GPIO_WritePin(GPIOA, GPIO_PIN_0, 0);
23            HAL_GPIO_WritePin(GPIOA, GPIO_PIN_1, 0);

```



Hình 7: Proteus Schematic for Exercise 4. Download project file at: [Excercise4/schematic/EX4.pdsprj](#)

```

24         HAL_GPIO_WritePin(GPIOA, GPIO_PIN_2, 1);
25         HAL_GPIO_WritePin(GPIOA, GPIO_PIN_3, 0);
26         HAL_GPIO_WritePin(GPIOA, GPIO_PIN_4, 0);
27         HAL_GPIO_WritePin(GPIOA, GPIO_PIN_5, 1);
28         HAL_GPIO_WritePin(GPIOA, GPIO_PIN_6, 0);
29         break;
30     case 3:
31         HAL_GPIO_WritePin(GPIOA, GPIO_PIN_0, 0);
32         HAL_GPIO_WritePin(GPIOA, GPIO_PIN_1, 0);
33         HAL_GPIO_WritePin(GPIOA, GPIO_PIN_2, 0);
34         HAL_GPIO_WritePin(GPIOA, GPIO_PIN_3, 0);
35         HAL_GPIO_WritePin(GPIOA, GPIO_PIN_4, 1);
36         HAL_GPIO_WritePin(GPIOA, GPIO_PIN_5, 1);
37         HAL_GPIO_WritePin(GPIOA, GPIO_PIN_6, 0);
38         break;
39     case 4:
40         HAL_GPIO_WritePin(GPIOA, GPIO_PIN_0, 1);
41         HAL_GPIO_WritePin(GPIOA, GPIO_PIN_1, 0);
42         HAL_GPIO_WritePin(GPIOA, GPIO_PIN_2, 0);
43         HAL_GPIO_WritePin(GPIOA, GPIO_PIN_3, 1);
44         HAL_GPIO_WritePin(GPIOA, GPIO_PIN_4, 1);
45         HAL_GPIO_WritePin(GPIOA, GPIO_PIN_5, 0);
46         HAL_GPIO_WritePin(GPIOA, GPIO_PIN_6, 0);
47         break;
48     case 5:
49         HAL_GPIO_WritePin(GPIOA, GPIO_PIN_0, 0);
50         HAL_GPIO_WritePin(GPIOA, GPIO_PIN_1, 1);
51         HAL_GPIO_WritePin(GPIOA, GPIO_PIN_2, 0);
52         HAL_GPIO_WritePin(GPIOA, GPIO_PIN_3, 0);
53         HAL_GPIO_WritePin(GPIOA, GPIO_PIN_4, 1);
54         HAL_GPIO_WritePin(GPIOA, GPIO_PIN_5, 0);
55         HAL_GPIO_WritePin(GPIOA, GPIO_PIN_6, 0);
56         break;
57     case 6:
58         HAL_GPIO_WritePin(GPIOA, GPIO_PIN_0, 0);
59         HAL_GPIO_WritePin(GPIOA, GPIO_PIN_1, 1);
60         HAL_GPIO_WritePin(GPIOA, GPIO_PIN_2, 0);
61         HAL_GPIO_WritePin(GPIOA, GPIO_PIN_3, 0);
62         HAL_GPIO_WritePin(GPIOA, GPIO_PIN_4, 0);
63         HAL_GPIO_WritePin(GPIOA, GPIO_PIN_5, 0);
64         HAL_GPIO_WritePin(GPIOA, GPIO_PIN_6, 0);
65         break;
66     case 7:
67         HAL_GPIO_WritePin(GPIOA, GPIO_PIN_0, 0);
68         HAL_GPIO_WritePin(GPIOA, GPIO_PIN_1, 0);
69         HAL_GPIO_WritePin(GPIOA, GPIO_PIN_2, 0);
70         HAL_GPIO_WritePin(GPIOA, GPIO_PIN_3, 1);
71         HAL_GPIO_WritePin(GPIOA, GPIO_PIN_4, 1);
72         HAL_GPIO_WritePin(GPIOA, GPIO_PIN_5, 1);

```

```

73         HAL_GPIO_WritePin(GPIOA, GPIO_PIN_6, 1);
74         break;
75     case 8:
76         HAL_GPIO_WritePin(GPIOA, GPIO_PIN_0, 0);
77         HAL_GPIO_WritePin(GPIOA, GPIO_PIN_1, 0);
78         HAL_GPIO_WritePin(GPIOA, GPIO_PIN_2, 0);
79         HAL_GPIO_WritePin(GPIOA, GPIO_PIN_3, 0);
80         HAL_GPIO_WritePin(GPIOA, GPIO_PIN_4, 0);
81         HAL_GPIO_WritePin(GPIOA, GPIO_PIN_5, 0);
82         HAL_GPIO_WritePin(GPIOA, GPIO_PIN_6, 0);
83         break;
84     case 9:
85         HAL_GPIO_WritePin(GPIOA, GPIO_PIN_0, 0);
86         HAL_GPIO_WritePin(GPIOA, GPIO_PIN_1, 0);
87         HAL_GPIO_WritePin(GPIOA, GPIO_PIN_2, 0);
88         HAL_GPIO_WritePin(GPIOA, GPIO_PIN_3, 0);
89         HAL_GPIO_WritePin(GPIOA, GPIO_PIN_4, 1);
90         HAL_GPIO_WritePin(GPIOA, GPIO_PIN_5, 0);
91         HAL_GPIO_WritePin(GPIOA, GPIO_PIN_6, 0);
92         break;
93     }
94 }

```

Program 5: Implementation of display_7SEG function

1.5 Exercise 5

Integrate the 7SEG-LED to the 4 way traffic light. In this case, the 7SEG-LED is used to display countdown value.

In this exercise, only source code is required to present. The function display7SEG in previous exercise can be re-used.

```

1 void led_traffic_with_7seg(int count) {
2     if (count >= 5) {
3         HAL_GPIO_WritePin(GPIOA, GPIO_PIN_8, 1);
4         HAL_GPIO_WritePin(GPIOA, GPIO_PIN_9, 1);
5         HAL_GPIO_WritePin(GPIOA, GPIO_PIN_10, 0);
6     } else if (count >= 3) {
7         HAL_GPIO_WritePin(GPIOA, GPIO_PIN_8, 1);
8         HAL_GPIO_WritePin(GPIOA, GPIO_PIN_9, 0);
9         HAL_GPIO_WritePin(GPIOA, GPIO_PIN_10, 1);
10    } else {
11        HAL_GPIO_WritePin(GPIOA, GPIO_PIN_8, 0);
12        HAL_GPIO_WritePin(GPIOA, GPIO_PIN_9, 1);
13        HAL_GPIO_WritePin(GPIOA, GPIO_PIN_10, 1);
14    }
15    HAL_Delay(100);
16 }

```

```

17
18 void led_traffic_with_7seg1(int count) {
19     if (count >= 5) {
20         HAL_GPIO_WritePin(GPIOA, GPIO_PIN_11, 0);
21         HAL_GPIO_WritePin(GPIOA, GPIO_PIN_12, 1);
22         HAL_GPIO_WritePin(GPIOA, GPIO_PIN_13, 1);
23     } else if (count >= 3) {
24         HAL_GPIO_WritePin(GPIOA, GPIO_PIN_11, 1);
25         HAL_GPIO_WritePin(GPIOA, GPIO_PIN_12, 1);
26         HAL_GPIO_WritePin(GPIOA, GPIO_PIN_13, 0);
27     } else {
28         HAL_GPIO_WritePin(GPIOA, GPIO_PIN_11, 1);
29         HAL_GPIO_WritePin(GPIOA, GPIO_PIN_12, 0);
30         HAL_GPIO_WritePin(GPIOA, GPIO_PIN_13, 1);
31     }
32     HAL_Delay(100);
33 }
34
35 // Inside while(1) loop:
36 while (1) {
37     if (counter < 0) {
38         counter = 9;
39     }
40     display_7SEG(counter--);
41     led_traffic_with_7seg(counter);
42     led_traffic_with_7seg1(counter);
43     HAL_Delay(1000);
44 }

```

Program 6: Traffic light helper functions and main loop integration for Exercise 5

1.6 Exercise 6

In this exercise, a new Proteus schematic is designed to simulate an analog clock, with 12 different number. The connections for 12 LEDs are supposed from PA4 to PA15 of the STM32. The arrangement of 12 LEDs is depicted as follows.

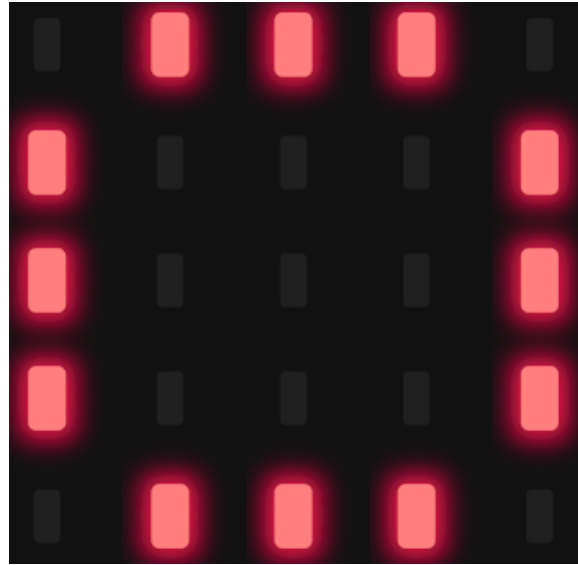
Report 1: Present the schematic.

Report 2: Implement a simple program to test the connection of every single LED. This testing program should turn every LED in a sequence.

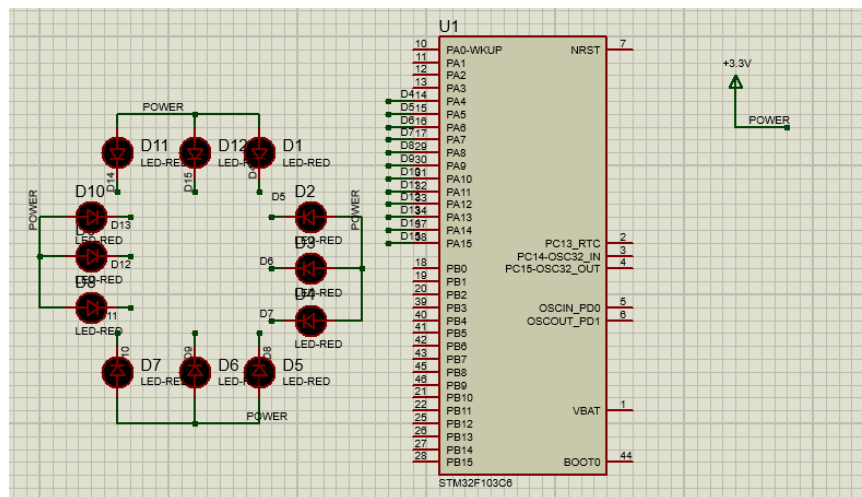
```

1 void leds_12() {
2     HAL_GPIO_TogglePin(GPIOA, GPIO_PIN_4); HAL_Delay(10);
3     HAL_GPIO_TogglePin(GPIOA, GPIO_PIN_5); HAL_Delay(10);
4     HAL_GPIO_TogglePin(GPIOA, GPIO_PIN_6); HAL_Delay(10);
5     HAL_GPIO_TogglePin(GPIOA, GPIO_PIN_7); HAL_Delay(10);
6     HAL_GPIO_TogglePin(GPIOA, GPIO_PIN_8); HAL_Delay(10);
7     HAL_GPIO_TogglePin(GPIOA, GPIO_PIN_9); HAL_Delay(10);
8     HAL_GPIO_TogglePin(GPIOA, GPIO_PIN_10); HAL_Delay(10);

```



Hình 8: 12 LEDs for an analog clock



Hình 9: Proteus Schematic for Exercise 6. Download project file at: Exercise6/schematic/EX6.pdsprj

```

9     HAL_GPIO_TogglePin(GPIOA, GPIO_PIN_11); HAL_Delay(10);
10    HAL_GPIO_TogglePin(GPIOA, GPIO_PIN_12); HAL_Delay(10);
11    HAL_GPIO_TogglePin(GPIOA, GPIO_PIN_13); HAL_Delay(10);
12    HAL_GPIO_TogglePin(GPIOA, GPIO_PIN_14); HAL_Delay(10);
13    HAL_GPIO_TogglePin(GPIOA, GPIO_PIN_15); HAL_Delay(10);
14    HAL_Delay(1000);
15 }
16
17 // Inside while(1) loop:
18 while (1) {
19     leds_12();
20 }

```

Program 7: LED testing function and usage in while loop

1.7 Exercise 7

Implement a function named **clearAllClock()** to turn off all 12 LEDs. Present the source code of this function.

```

1 void clearAllClock(void) {
2     HAL_GPIO_WritePin(
3         GPIOA, GPIO_PIN_4 | GPIO_PIN_5 | GPIO_PIN_6 |
4         GPIO_PIN_7 |
5         GPIO_PIN_8 | GPIO_PIN_9 | GPIO_PIN_10 |
6         GPIO_PIN_11 |
7         GPIO_PIN_12 | GPIO_PIN_13 | GPIO_PIN_14 |
8         GPIO_PIN_15,
9         GPIO_PIN_SET
10    );
11 }

```

Program 8: Implementation of clearAllClock function

1.8 Exercise 8

Implement a function named **setNumberOnClock(int num)**. The input for this function is from **0 to 11** and an appropriate LED is turn on. Present the source code of this function.

```

1 void setNumberOnClock(int num) {
2     if (num < 0 || num > 11)
3         return;
4     // Turn on the LED corresponding to index num (active
5     // low)
6     HAL_GPIO_WritePin(GPIOA, (1 << (num + 4)),
7         GPIO_PIN_RESET);
8 }

```

Program 9: Implementation of setNumberOnClock function

1.9 Exercise 9

Implement a function named **clearNumberOnClock(int num)**. The input for this function is from **0 to 11** and an appropriate LED is turn off.

```
1 void clearNumberOnClock(int num) {
2     if (num < 0 || num > 11)
3         return;
4     // Turn off the LED corresponding to index num (active
5     low)
6     HAL_GPIO_WritePin(GPIOA, (1 << (num + 4)), GPIO_PIN_SET
7 );
8 }
```

Program 10: Implementation of clearNumberOnClock function

1.10 Exercise 10

Integrate the whole system and use 12 LEDs to display a clock. At a given time, there are only 3 LEDs are turn on for hour, minute and second information.

```
1 void display_clock(int hour, int minute, int second) {
2     minute += second / 60;
3     second %= 60;
4     hour += minute / 60;
5     minute %= 60;
6
7     clearAllClock();
8     setNumberOnClock(hour % 12);
9     setNumberOnClock((int) minute / 5);
10    setNumberOnClock((int) second / 5);
11 }
12
13 // Inside while(1) loop:
14 while (1) {
15     display_clock(10, 20, 20);
16     HAL_Delay(1000);
17 }
```

Program 11: Implementation of display_clock function and main application execution